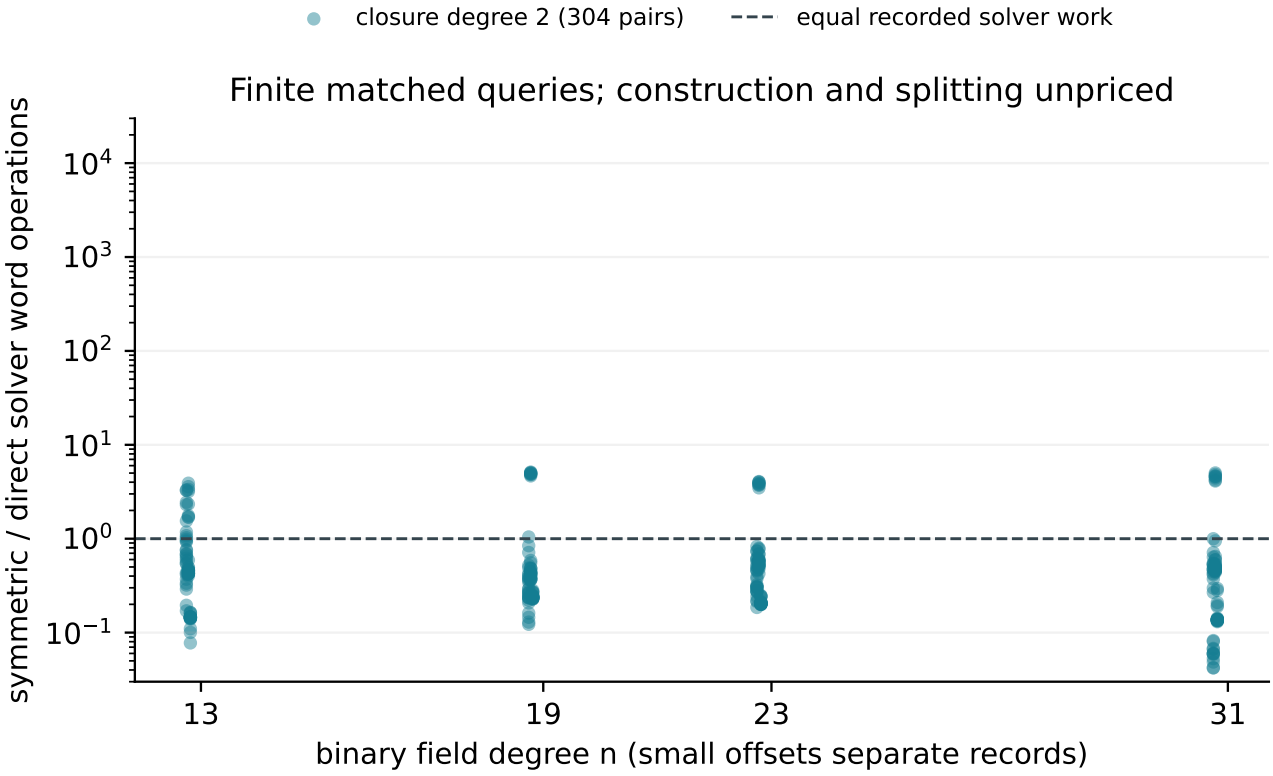
Structural closure panel: degree 2

Run: `structural-d2-20261023-01`. Experiments 2026-10-23; report 2026-10-25.

The frozen panel has 96 commands and 1152 rows: $n = 13, 19, 23, 31$, $\ell = 2, 3, 4$, prefix/geometric/trace-zero geometric/random bases, two recorded base replicates and four public target draws. Prefix replicate 1 is a duplicate; 144 rows are excluded from the primary table. Three formulations run: chained S3, direct S4, symmetric product-space S4.

Limits: degree-2 closure, 30,000 columns, 12 GiB address space, 900 seconds per command. This shared host earns L0: timings are diagnostic only. The hybrid extraction has an input-degree floor, which reaches six for direct S4. These are not equal-degree or whole-query speed measurements.

There are 304 completed direct/symmetric pairs among 336 distinct input queries. The remaining 32 are unpaired, not zero-cost. Symmetric solver work is lower in 263 pairs, higher in 41, equal in 0. Recorded ratios range 0.04201 to 5.123. Independent direct point enumeration checks every completed answer; the audit reports 768 complete extractions and zero failures.



For the completed symmetric queries, 8100 algebraic roots produced 4 verified x-multisets; 8096 roots failed splitting or signed lifting. Construction, splitting and physical conversion are not priced in solver words. No end-to-end DLP cost, rho ratio, scaling law or external-CAS speed claim is made.

Raw rows: `../../runs/structural-d2-20261023-01/cell-*/rows.jsonl`. The native recomputation is `examples/ic_campaign_report_data.rs`; plot values are `../../report-data.json`. The frozen protocol, controls, input/binary hashes, cap refusals and setup failures remain in the study.